these minutes, transcripts, and other materials to describe Fed behavior,[9] an important alternative is a reduced-form model developed by the Stanford University economist John Taylor in 1993.

The famous *Taylor rule* states that the Fed funds rate should be set to move together with inflation and economic activity. Its original formulation was

$$FF_t = r^* + \pi_t + 0.5(\pi_t - \pi^*) + 0.5Gap_t, \tag{9.1}$$

where FF denotes the Fed funds rate, r^* is the long-run real interest rate, π is the current inflation rate, and π^* is the target inflation rate. The output gap, which is the difference between real and potential GDP, is denoted as *Gap*. The Taylor rule is used both as a descriptive tool (what has the Fed done?) and a prescriptive tool (what should the Fed do?).

The Taylor rule captures the dual mandate of the Fed by describing how it moves the Fed funds rate in response to changes in the output gap and inflation. There is a positive coefficient of 0.5 (the coefficient differs in data and subsequent theories) on the output gap. As the output gap shrinks and economic activity slows, the Fed lowers the Fed funds rate to spur economic growth. The coefficient on inflation, π, in equation (9.1) is 1.5, but it is split up between the first term, $r^* + \pi$, and the second, $0.5(\pi - \pi^*)$. The first term is the sum of the real rate and inflation and should be the simple nominal short rate according to the *Fisher Hypothesis* (see below). The second term is the deviation of current inflation from its long-term target. As inflation increases, the Fed raises the Fed funds rate to rein in rising prices. As the total coefficient on inflation is greater than 1.0, when inflation increases, the Fed moves interest rates more than one for one. This means that the Fed raises *real rates* when inflation increases. This is called the *Taylor principle*.

Many versions of the Taylor rule describe the Fed's reaction to inflation and economic growth. These models can use different definitions of inflation (GDP deflator, CPI-All, core CPI, PPI, etc.) or of economic growth (GDP growth, output gap, industrial production growth, employment, etc.); the important concept in equation (9.1) is that the Fed responds to general movements in inflation and growth. Variants developed since Taylor (1993) can use forward-looking measures like surveys or agents' expectations (*forward-looking rules*) and past data beyond just current inflation and output (*backward-looking rules*). Versions of the rule can also account for occasions when the Fed partially adjusts to information (*partial adjustment rules*).[10] The difference between the actual Fed funds rate and what the

[9] See, for example, the recent study of Weise (2012), who examines the transcripts to characterize Fed behavior during the 1970s. Romer and Romer (2000) show that the Fed's internal forecasts are far superior to commercial forecasts.

[10] There is an equivalence among some forms of these different basic Taylor rules, and the forward-looking, backward-looking, and partial adjustment versions, as Ang, Dong, and Piazzesi (2007) explore.